ONE-PAGE BRIEF

# Post-quantum cryptography review: rust-secp256k1 @ fe8b3bec962f

Client: rust-secp256k1 — public repository (passive review) · Source: github.com/rust-bitcoin/rust-secp256k1, commit `fe8b3bec962f` · Reviewed: 2026-11-04 19:19 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

**Level 0 of 4**

no post-quantum cryptography

FINDINGS

**0** Critical **0** High **5** Medium

**0** Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | | |
|---|---|---|
| Medium | QSI-2026-12584 | **Quantum-vulnerable ECDSA: transaction and message signing (23 file(s))** preliminary<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium | QSI-2026-12586 | **Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (3 file(s))** preliminary<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium | QSI-2026-12585 | **Quantum-vulnerable ECDH: role not determined (5 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in... |
| Medium | QSI-2026-12587 | **Quantum-vulnerable ECDSA: role not determined (4 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form... |

1 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium transaction and message signing | ECDSA, Schnorr (BIP-340) | Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum... |
| Medium role not determined | ECDH, ECDSA, Schnorr (BIP-340) | Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS... |

Static review of source code and manifests; files analysed: 118. No requests were sent to the owner's systems.

Full report, content SHA-256: `82abeea34e02e233b7a86c4ef175f23ed70a059aed003f355163c56f669bdd1b`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.